CNSCC.160 Lab 5 Speech Endpoint Detection

The goal is to separate "Speech" from "Silence/Background Noise" by analysing the signal in short frames.

1. Background information

1.1 Short-Time Energy (STE)

Voiced speech (vowels) has high amplitude, while silence has very low energy.

- **Algorithm:** Divide the signal into overlapping frames. For each frame i, calculate:

$$E_i = \sum_{n=1}^{N} [x(n)]^2$$

(where N is the frame length).

- **Notes:** Use `buffer` to segment the signal and `sum(frame.^2)` to find the energy profile.

1.2 Zero Crossing Rate (ZCR)

This identifies "Unvoiced" speech (like /s/ or /f/) which looks like high-frequency noise.

- **Algorithm:** Count how many times the signal crosses the zero axis within a frame.

$$ZCR = \frac{1}{2N} \sum_{n=1}^{N} |sgn[x(n)] - sgn[x(n-1)]|$$

- **Notes:** Use diff(sign(frame) ~= 0) to find transitions.

2. Lab Guidelines

Step 1: Signal Pre-processing

- **Normalization:** Normalizing to **1.0** makes the thresholding math consistent across different microphones or recordings.

Step 2: Framing

- **The buffer Function:** Understand Matlab buffer function. We process speech in "frames" (usually **20–30ms**) because speech is "quasi-stationary" (its properties change over time, but stay stable for very short bursts).

- **Overlap:** Use 50% overlap to ensures that a loud event happening at the very edge of a frame isn't "split" and missed by the detector.

Step 3: Feature Extraction through STE or ZCR

- **Mathematical Definition:** Implement the formula shown above.

Note, we are doing this to frames.

Step 4: Adaptive Thresholding Methods

- Try hard thresholding first.
- **The "Adaptive" Solution:** Instead of a hard number (like 0.1), use the first **N frames** to measure the environment. The formula `Mean + k * StdDev` allows the detector to adjust itself to the room's noise floor.

Step 5: Mask Refinement (VAD Logic)

- **Median Filtering:** Human speech has "dips" in energy during stop consonants (like 'p' or 'k'). The median filter acts as a "bridge" to keep the detector active during these tiny, natural pauses.

3. Lab Tasks

Complete the following tasks using both Harvard_sentences.wav and moon.wav files.

Task 3.1: Pre-processing & Framing

- Load the audio, normalize the amplitude, and remove the DC offset.
- Segment the signal into 25ms frames using the buffer function with 50% overlap.

Task 3.2: Short-Time Energy (STE) Analysis

- Calculate the STE for every frame.
- Implement an adaptive threshold based on the first 10 frames of silence.
- **Plotting:** Create Figure 1 showing the Signal, STE curve and threshold line, and the Resulting Mask in 3 subplots.

Task 3.3: Zero Crossing Rate (ZCR) Analysis

- Calculate the rate of sign changes in each frame.
- Implement a similar adaptive threshold to distinguish between background hiss and unvoiced speech (consonants).
- **Plotting:** Create Figure 2 showing the Signal, ZCR curve and threshold line, and the Resulting Mask in 3 subplots.

Task 3.4: Combined Feature Endpoint Detection

- Use a logical OR (|) to combine the STE and ZCR masks.

- Apply a **Median Filter** to remove "chatter" (short noise spikes) and fill syllabic gaps.
- Identify the absolute **Start** and **End** times of the utterance.
- **Plotting:** Create Figure 3 with markers for the final detected endpoints.

4. Submission Requirements

You should submit a single PDF report and a Matlab file. The report should have a title, name and student ID in the front page. Then sections like: Objective, Theory, Method/Procedure, Results and Analysis. Remember to include the following:

1. Annotated Figures for both audio files.

- **Three Figure Plots:** High-resolution exports of the three figures generated (STE-only, ZCR-only, and Combined).
- **Observations:** For each figure, you must identify one instance of a **voiced** sound (high STE) and one instance of an **unvoiced** sound (high ZCR) in the waveform.

2. Code Implementation

- **Script File (.m):** A clean, commented version of their MATLAB script.
- **Threshold Discussion:** A brief explanation of why they chose their specific multipliers (k values) for the adaptive thresholds, especially regarding the noisy "Moon.wav" file.

3. Analysis Questions

- Describe how the logical OR operation and the Median Filter improved the detection accuracy.

5. Submission Deadlines

Submission deadline is 8th May 2026.